

# INDIVIDUAL LOGBOOK

## CSE3433 SOFTWARE ARCHITECTURE

Semester II 2025/2026

### STUDENT INFORMATION

**Student Name:** Muhammad Adam Danish bin Mohamad Faiz

**Matric Number:** S75851

**Group Number:** Group 15

**Project Title:** Event-Driven Job Recruitment & Application Processing System

**Role:** Full Stack & Architecture Lead

**Lecturer:** Dr Mohamad Nor

### TEAM MEMBER AND ROLES

| Name                                  | Matric Number | Role                           |
|---------------------------------------|---------------|--------------------------------|
| Muhammad Radilan bin Yaakub           | S75736        | Backend & Deployment Lead      |
| Muhammad Adam Danish bin Mohamad Faiz | S75851        | Full Stack & Architecture Lead |

# WEEKLY PROGRESS

## Week 3 - Project Analysis and Proposal Development

| Weekly Progress Matric   | Details & Data Logs                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tasks Undertaken         | Conducted a detailed analysis of the recruitment process and identified common issues faced by traditional hiring systems. Discussions were held with team members to determine the project scope, target users, and expected system functionalities. Research was also carried out on existing online recruitment platforms to understand current industry practices and limitations. |
| Individual Contributions | Prepared the initial problem description by identifying inefficiencies such as delayed application processing, poor communication between recruiters and applicants, and limited scalability of traditional systems. Contributed ideas on how an Event-Driven Architecture (EDA) approach could improve responsiveness and automation within the recruitment process.                  |
| Challenges Faced         | The main challenge was narrowing down the project scope to ensure the proposed system could be completed within the semester timeline while still meeting all project requirements.                                                                                                                                                                                                    |
| Action Plan              | Finalize the project objectives and begin documenting the system requirements for the proposal submission.                                                                                                                                                                                                                                                                             |

## **Week 5 – Deliverable 1: Project Proposal**

| <b>Weekly Progress Matric</b>   | <b>Details &amp; Data Logs</b>                                                                                                                                                                                                 |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Tasks Undertaken</b>         | Completed the Project Proposal document and reviewed all sections before submission. Team discussions were conducted to verify that the proposed solution aligned with project requirements and architecture objectives.       |
| <b>Individual Contributions</b> | Prepared the Event-Driven Architecture explanation, contributed to the functional requirements section, and assisted in refining the project objectives. Performed final proofreading and formatting of the proposal document. |
| <b>Challenges Faced</b>         | Ensuring all proposal sections were clearly connected and adequately justified within the limited page requirement.                                                                                                            |
| <b>Action Plan</b>              | Begin detailed architecture design and system modelling activities.                                                                                                                                                            |

## **Week 7 – Deliverable 2: Architecture Design**

| <b>Weekly Progress Matric</b>   | <b>Details &amp; Data Logs</b>                                                                                                                                                                                                    |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Tasks Undertaken</b>         | Completed all architecture design documentation and diagrams required for submission. The architecture design was reviewed to ensure consistency with the proposed Event-Driven Architecture approach.                            |
| <b>Individual Contributions</b> | Developed the System Context Diagram, Component Architecture Diagram, Interaction Diagram, and Deployment Diagram. Documented non-functional requirements such as scalability, responsiveness, availability, and maintainability. |
| <b>Challenges Faced</b>         | Representing asynchronous communication flows clearly in the interaction and component diagrams.                                                                                                                                  |
| <b>Action Plan</b>              | Transition from design activities to prototype planning and implementation.                                                                                                                                                       |

## **Week 8 – Prototype Development**

| <b>Weekly Progress Matric</b>   | <b>Details &amp; Data Logs</b>                                                                                                                                                                                                        |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Tasks Undertaken</b>         | Planned the implementation strategy and discussed the technology stack to be used for development. Team members reviewed system modules and development responsibilities.                                                             |
| <b>Individual Contributions</b> | Defined the major events used within the system, including Application Submitted, Application Reviewed, Job Created, and Interview Scheduled events. Assisted in mapping architecture components to practical implementation modules. |
| <b>Challenges Faced</b>         | Ensuring that the implementation plan accurately reflected the architecture design produced earlier.                                                                                                                                  |
| <b>Action Plan</b>              | Begin development and integration of core system modules.                                                                                                                                                                             |

## Week 9

| Weekly Progress Matric   | Details & Data Logs                                                                                                                                                                                                                         |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tasks Undertaken         | Monitored prototype development progress and reviewed the implementation of major system modules. Team discussions focused on integrating frontend and backend functionalities.                                                             |
| Individual Contributions | Reviewed candidate application workflows and verified that implementation activities remained aligned with architecture specifications. Assisted in refining dashboard structures and navigation flows for both recruiters and job seekers. |
| Challenges Faced         | Maintaining consistency between architecture design documentation and actual system implementation.                                                                                                                                         |
| Action Plan              | Continue development and begin preliminary testing activities.                                                                                                                                                                              |

## Week 10

| Weekly Progress Matric          | Details & Data Logs                                                                                                                                                                 |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Tasks Undertaken</b>         | Performed testing on completed modules and reviewed user interactions within the system. Various workflows were evaluated to identify functional issues.                            |
| <b>Individual Contributions</b> | Tested application submission processes, user authentication functionality, and job management features. Documented identified issues and provided recommendations for improvement. |
| <b>Challenges Faced</b>         | Resolving integration issues between user interfaces, database operations, and backend processing logic.                                                                            |
| <b>Action Plan</b>              | Finalize the prototype and prepare demonstration materials for Deliverable 3.                                                                                                       |

### **Week 11 – Deliverable 3: Prototype Progress**

| Weekly Progress Matric          | Details & Data Logs                                                                                                                                                                                            |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Tasks Undertaken</b>         | Completed and demonstrated the initial working prototype. Screenshots and implementation evidence were prepared for submission together with the project repository.                                           |
| <b>Individual Contributions</b> | Verified the functionality of login, job posting, application submission, job search, and application tracking modules. Assisted in documenting implementation progress and preparing demonstration materials. |
| <b>Challenges Faced</b>         | Ensuring all modules operated correctly and consistently during the demonstration process.                                                                                                                     |
| <b>Action Plan</b>              | Conduct architecture evaluation and system performance review.                                                                                                                                                 |



## **Week 12**

| <b>Weekly Progress Matric</b>   | <b>Details &amp; Data Logs</b>                                                                                                                                                                       |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Tasks Undertaken</b>         | Evaluated the implemented system against the architecture objectives and quality requirements established earlier in the project. Reviewed system performance and maintainability aspects.           |
| <b>Individual Contributions</b> | Analyzed how Event-Driven Architecture contributed to scalability, modularity, and responsiveness. Documented architecture evaluation findings and prepared supporting content for the final report. |
| <b>Challenges Faced</b>         | Evaluating scalability and performance without access to a large-scale deployment environment.                                                                                                       |
| <b>Action Plan</b>              | Prepare and compile the final project report.                                                                                                                                                        |

### **Week 13 – Deliverable 4: Project Report and Individual Logbook**

| <b>Weekly Progress Matrix</b>   | <b>Details &amp; Data Logs</b>                                                                                                                                                                                                               |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Tasks Undertaken</b>         | Completed the final project report and individual logbook documentation. Conducted final reviews of all deliverables to ensure consistency and completeness.                                                                                 |
| <b>Individual Contributions</b> | Prepared the Architecture Evaluation section, Team Contributions section, and Conclusion and Reflection section. Verified that the final report accurately reflected the proposal, architecture design, and prototype implementation stages. |
| <b>Challenges Faced</b>         | Combining all project deliverables into a comprehensive report while maintaining consistency across documentation.                                                                                                                           |
| <b>Action Plan</b>              | Submit the final report and reflect on the overall project experience and lessons learned.                                                                                                                                                   |

## GENERATIVE AI USAGE RECORD

| Week    | Tool Used | Purpose                 | Prompt Used                                        | Output Summary                     | How It was Used                       |
|---------|-----------|-------------------------|----------------------------------------------------|------------------------------------|---------------------------------------|
| Week 3  | ChatGPT   | Problem Analysis        | Identify issues in traditional recruitment systems | Generated problem analysis points  | Used for problem description research |
| Week 4  | ChatGPT   | Architecture Research   | Explain Event-Driven Architecture and benefits     | Provided architecture explanations | Used for architecture selection       |
| Week 6  | ChatGPT   | Service Identification  | Suggest services for recruitment system using EDA  | Generated component suggestions    | Assisted architecture planning        |
| Week 7  | ChatGPT   | Documentation Support   | Explain non-functional requirements for EDA        | Generated requirement examples     | Used in architecture document         |
| Week 9  | ChatGPT   | Development Assistance  | Suggest recruitment system workflow                | Generated workflow recommendations | Used to verify implementation flow    |
| Week 10 | ChatGPT   | Testing Support         | Generate software testing checklist                | Produced testing ideas             | Used during prototype evaluation      |
| Week 12 | ChatGPT   | Architecture Evaluation | Evaluate advantages of EDA                         | Generated evaluation points        | Used in evaluation section            |
| Week 13 | ChatGPT   | Report Writing          | Help write conclusion and reflection               | Generated draft content            | Adapted into final report             |